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Atlantic Technological University

Practical No: 3

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Title: Sampling, Quantisation and DAC Reconstruction

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Course: Bachelor of Engineering in Electronic Engineering

Subject: Digital Communications and Transmission

Lecturer: Ben Graham

Summary:

This practical investigates the design and implementation of a 4-bit Pulse Code Modulation system using an MSP430 microcontroller.

Objective:

The primary aim of this lab was to build a 4-bit Pulse Code Modulation (PCM) system capable of digitising a sine wave and later reconstructing it through a discrete summing amplifier. Doing so helped bridge the gap between theory and practice, particularly by setting up hardware timers on the MSP430 microcontroller to handle the sampling routines. The raw 10-bit data from the ADC was then mapped and downsampled into a 4-bit parallel output, effectively covering both the quantisation and binary encoding stages. Once the digital data was processed, the circuit successfully rebuilt the analogue waveform. This allowed for a practical evaluation of the system's physical limits, including visible quantisation noise, voltage scaling constraints, and the distortion caused by aliasing.

Theory:
Pulse Code Modulation, PCM, fundamentally relies on three core stages: sampling, quantisation, and encoding. In this experiment, the microcontroller's hardware timer was set up to trigger the ADC, capturing the analogue waveform at a precise 20 Hz sampling rate. The raw 10-bit data generated by the ADC was then scaled down to 4 bits using software bitwise operations. This digital downsampling effectively quantised the signal into 16 discrete voltage levels and encoded the resulting binary values for transmission across the parallel output pins.

Equipment Used:
Signal Generator – MP750510
Oscilloscope – Tektronix TDS 2012B
DC Power Supply – Thandar EL302Tv
MSPG2553 Development Board
741 Op-Amp
Resistors:

- 5k resistors (1)
- 10k resistors (4)
- 20k resistor (1)
- 40k resistors (1)
- 80k resistors (1)

1. Analogue Input Generation

To begin the experiment, a 2 Hz sinusoidal signal was created using the MP750510 generator. The amplitude and DC offset were adjusted to ensure the voltage stayed perfectly within the 0V to 3V limits. The specific generator settings can be seen in Figure 1, and the physical test of this waveform captured by the oscilloscope is presented in Figure 2.

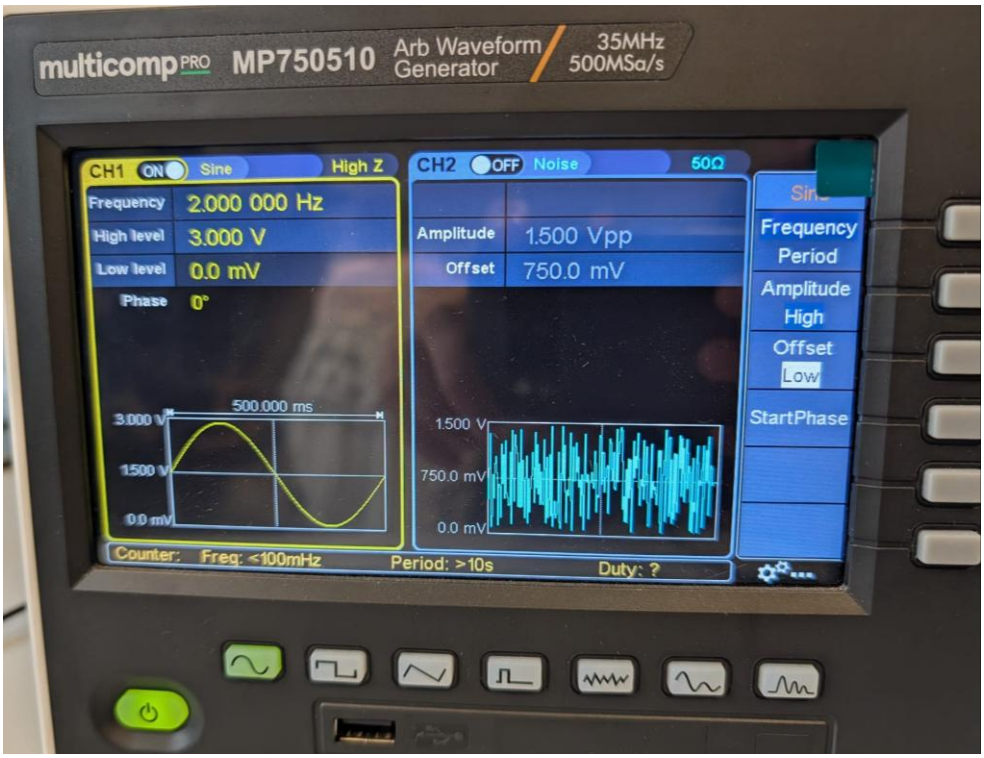


Figure 1: Configured wave

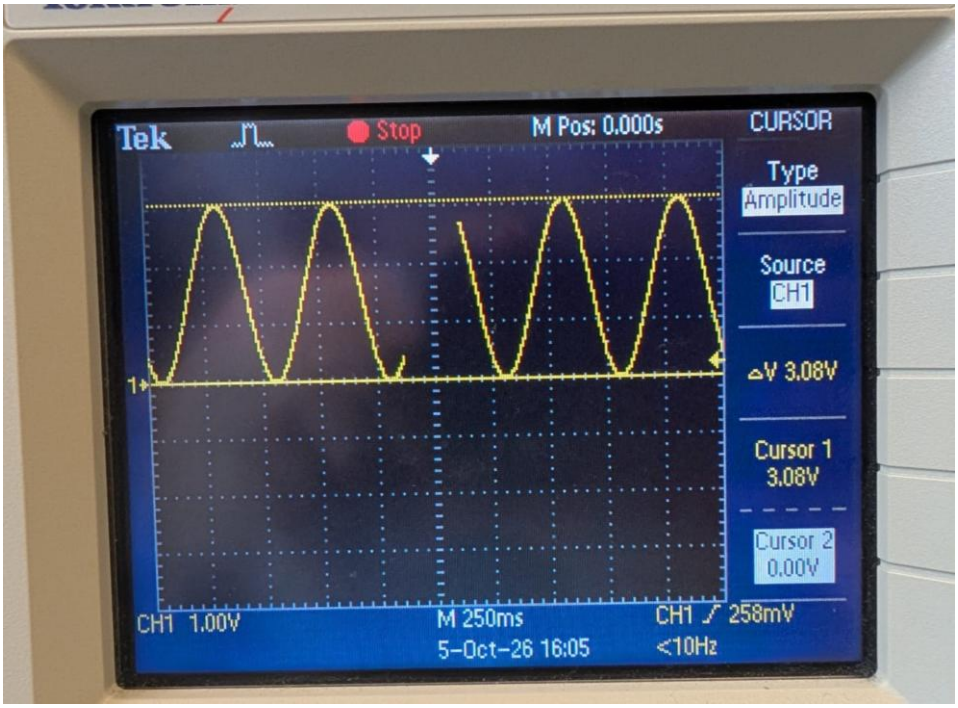


Figure 2: Waveform verified in the Oscilloscope

2. Hardware Configuration & Sampling Routine

During this phase of the practice, the necessary hardware configuration and sampling routines were established to process the generated signal.
Initially, the microcontroller's pins P1.0, P1.1, P1.2, and P1.3 were configured as outputs, while pin P1.4 was exclusively designated as the input for the ADC. This setup allows the digital data to be transmitted in a 4-bit parallel form.
Furthermore, the 10-bit ADC was initialized and configured to accurately capture the incoming analogue signal on pin P1.4.
Finally, the internal hardware, Timer A, was configured to establish a precise sampling rate of 20 Hz.

main.c

```c
#include <msp430.h>

void init_timer(void) {
    // Retaining your requested 20 Hz sample rate
    TA0CCR0 = 50000;                    // 1MHz / 50000 = 20 Hz sample rate
    TA0CCTL0 = CCIE;
    TA0CTL = TASSEL_2 + MC_1 + TACLR;
}
void init_adc(void) {
    ADC10CTL0 = SREF_0 + ADC10SHT_3 + ADC10ON + ADC10IE;
    // Example ADC10 initialization - make sure ADC10IE is included!
    ADC10CTL1 = INCH_4 + SHS_0 + ADC10SSEL_3; // A4 Input (P1.4)

    ADC10AE0 |= BIT4;
}

volatile unsigned int adc_raw = 0;
volatile unsigned char pcm_4bit = 0;

int main(void) {
    WDTCTL = WDTPW | WDTHOLD;   // Stop watchdog timer

    // Configure Clock to 1 MHz
    BCSCTL1 = CALBC1_1MHZ;
    DCOCTL = CALDCO_1MHZ;

    // Configure Outputs: P1.0 (LSB), P1.1, P1.2, P1.3 (MSB)
    P1DIR |= (BIT0 | BIT1 | BIT2 | BIT3);
    // Disable internal pull-up/pull-down resistors on Pins 0, 1, 2, 3
    P1REN &= ~(BIT0 | BIT1 | BIT2 | BIT3);
    P1OUT &= ~(BIT0 | BIT1 | BIT2 | BIT3);

    init_adc();
    init_timer();

    __enable_interrupt();       // Enable global interrupts

    while(1) {
        __bis_SR_register(LPM0_bits); // Sleep in low power mode
    }
}
```

Figure 3: Code in Code Composer Studio

3. Quantization and Encoding of the signal:

Using the code described in the previous paragraph, different voltages (0, 1, 2, and 3 V) were tested, thereby verifying that the theoretical values calculated in Figure 4 were correct. For this purpose, the signal from the signal generator was connected to pin P1.4 of the MSP430. Once processed, the microcontroller outputs the corresponding numerical value in binary through pins P1.0 - P1.3, with P1.0 being the LSB and P1.3 the MSB. One final interesting detail learned during this part of the practical is that using the MP750510 signal generator does

not yield consistent results due to how this device operates, as its signal is not constant. On the other hand, the EL302Tv signal generator did allow for generating the signal correctly, as it generates a constant signal.

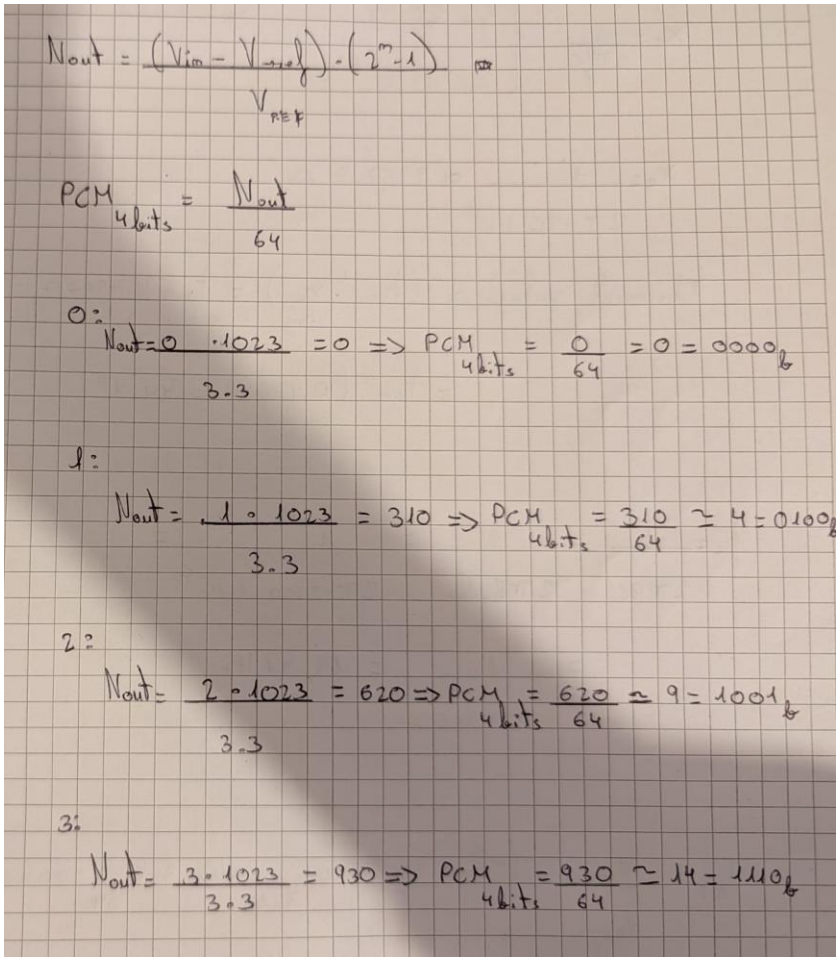


Figure 4: Calculation of theoretical values

To sum up:

Input Voltage (V)	4-bit Binary Output
0V	0000
1V	0100
2V	1001
3V	1110

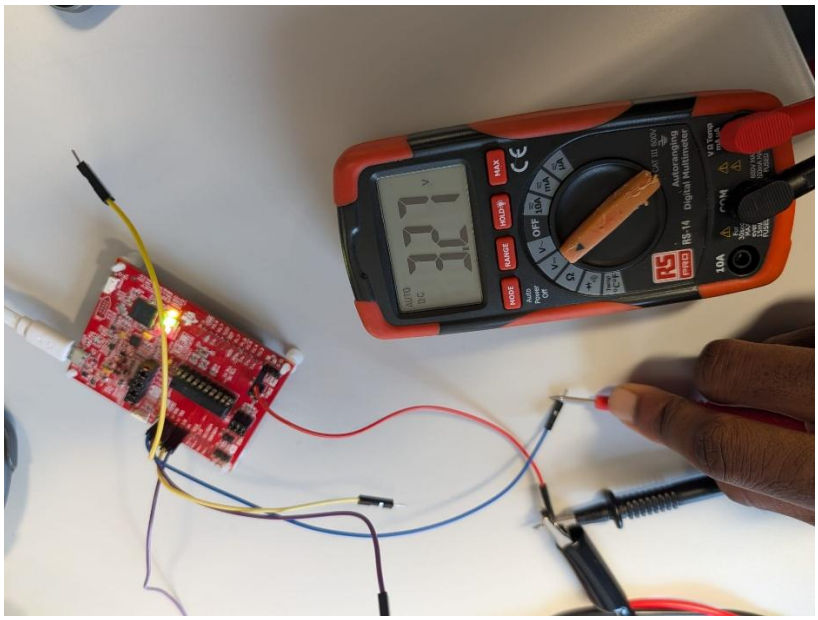


Figure 5: Measure of P1.0 with 3.3V in P1.4

4. Discrete DAC Hardware Reconstruction

In the final part of the practical, the digital data had to be turned back into an analogue waveform like the original input. A 4-channel summing amplifier was set up on a breadboard with UA741 op-amps to act as the DAC.

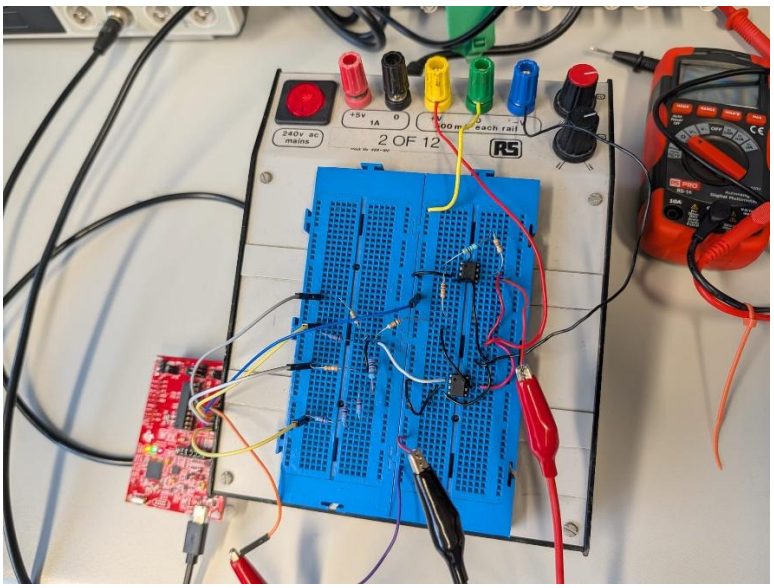


Figure 6: Summing Amplifier

During testing, the reconstructed signal was tested on the oscilloscope, as seen in Figure 7. While the wave clearly had the staircase look typical of quantisation, confirming the circuit's basic operation, it was missing the full 16 voltage steps of a 4-bit setup. Instead, the voltage jumps were quite large and there were

noticeably fewer steps. A poor physical connection in the breadboard, or perhaps a damaged component, probably caused this drop in resolution, making the DAC behave more like a 3-bit (8-level) system.

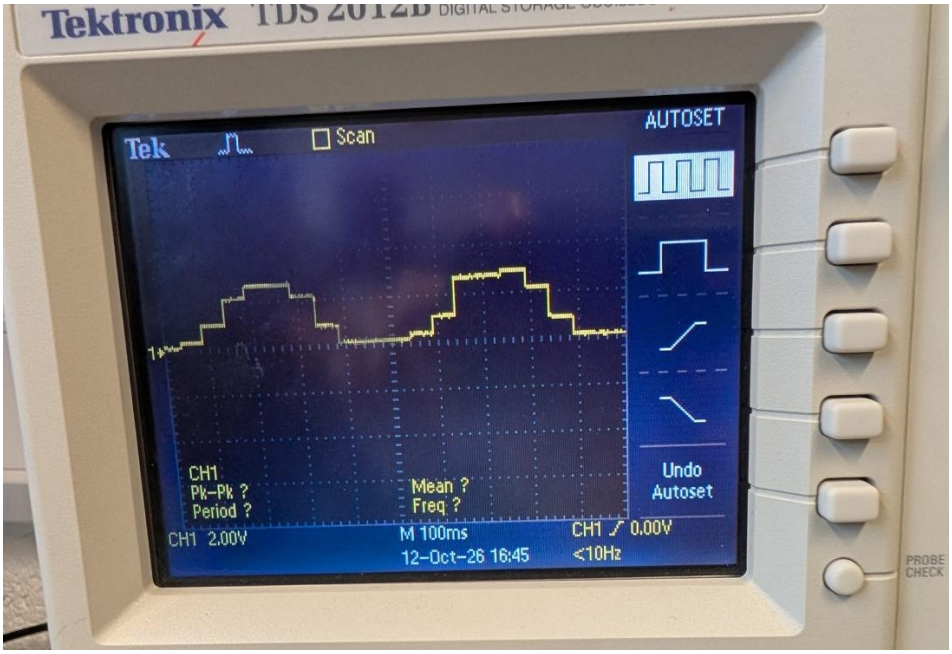


Figure 7: Reconstructed Signal

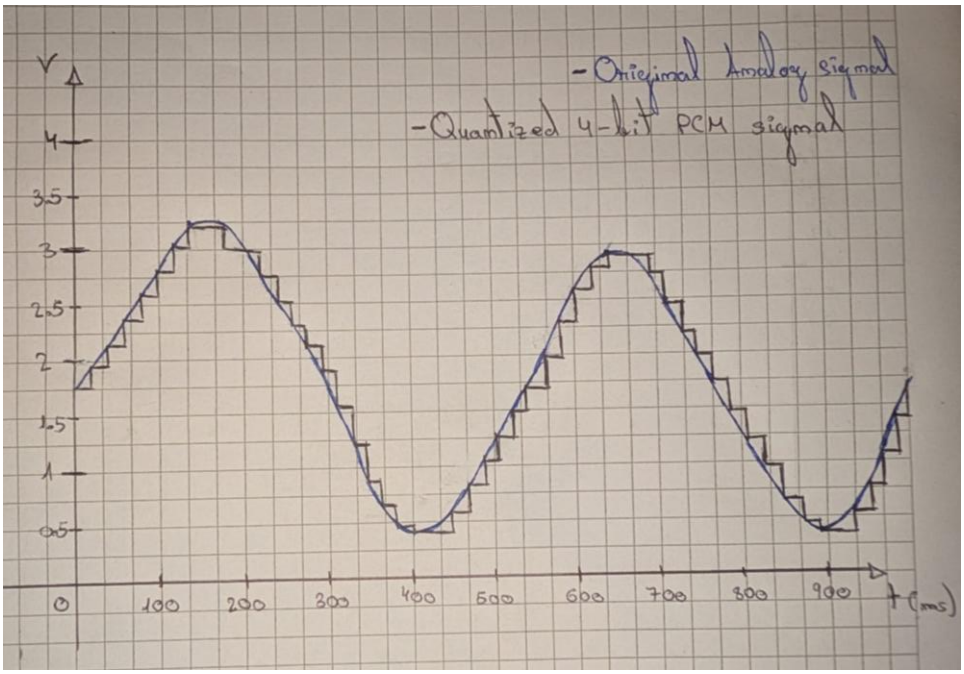


Figure 8: Quantized 4-bit PCM Signal drawn

5. Analysis and Questions

Question 1: What would the output be for binary value of 1,1,1,1

As shown in Figure 9, the corresponding Vout would be -6.1875 V.

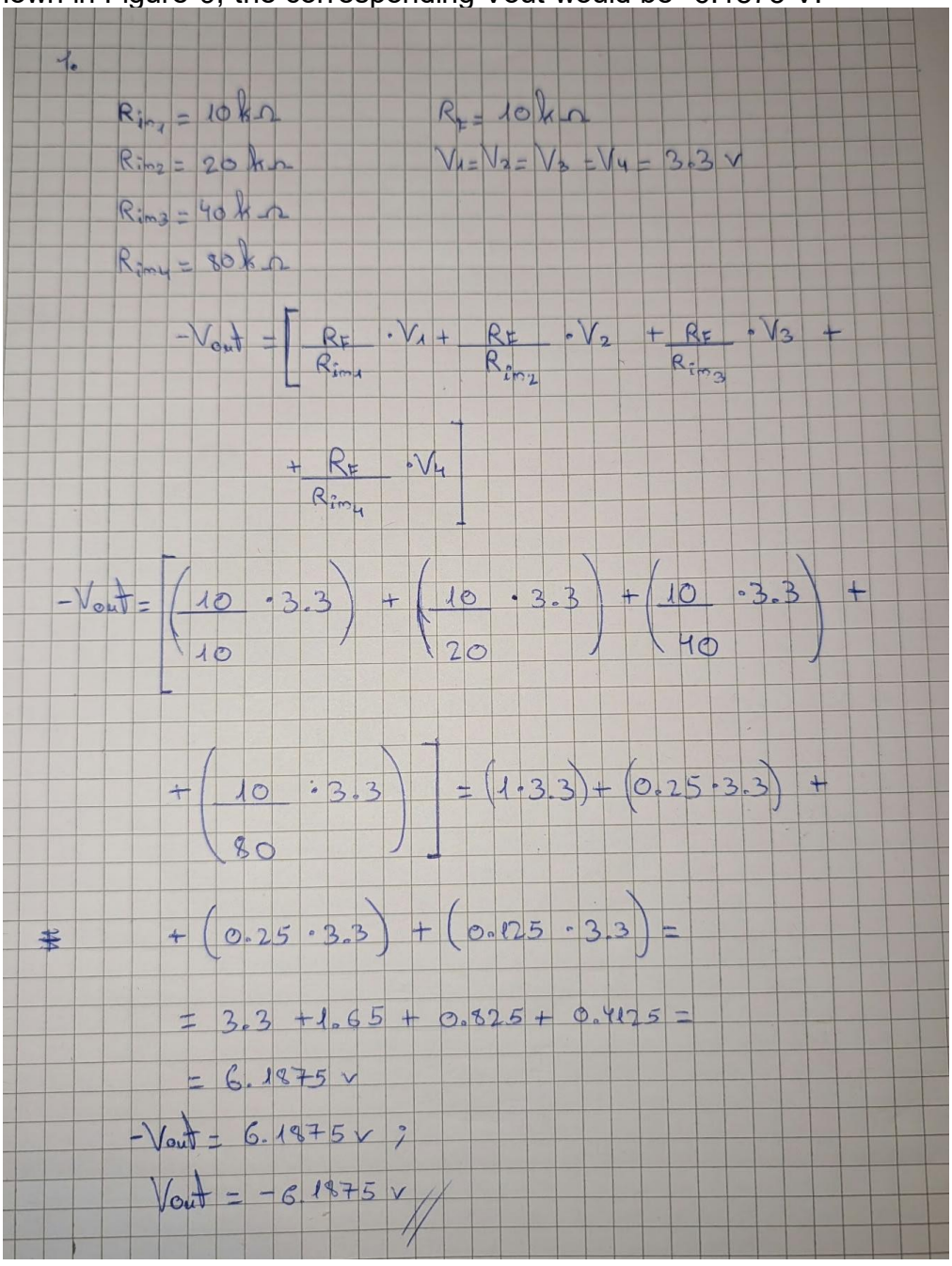


Figure 9: Calculation binary value of 1111

Question 2: Explain the purpose of the second op-amp in the circuit, note the resistor values which control the gain of the second op-amp will need to changed.
Remember the maximum voltage output from the DAC must be 3V for binary 1,1,1,1.

The first stage of the circuit uses a summing amplifier, U1, that combines the four digital bits while inverting their polarity, delivering -6.1875 V when the binary input is at its maximum (1111). Because of this, the second operational amplifier ,U2, is required to perform two essential tasks: polarity inversion and amplitude scaling.
Configured as an inverting amplifier, U2 flips the negative output from U1 back into a positive voltage range. At the same time, its gain is controlled by the ratio between its feedback resistor Rf and its input resistor Rin. Since both resistors are currently set to 10kΩ in the initial schematic, the stage has a unity gain of -1, resulting in +6.1875 V. Therefore, the resistor values controlling this gain must be modified to attenuate the signal so that the maximum output never exceeds the required +3.0 V limit for a 1,1,1,1 input.

Question 3: Determine the precise scaling factor required and calculate the exact replacement value for the feedback resistor Rf on the second amplifier to achieve this.

To step down the 6.1875 V peak from the first stage to the required 3.0 V maximum limit, the second op-amp stage needs a specific attenuation factor. This precise scaling factor is found by dividing the target output voltage by the incoming voltage from U1.
Since the voltage gain of the second inverting op-amp is determined by the ratio Gain = Rf / Rin, and the input resistor Rin connecting U1 to U2 is kept at 10kΩ, the exact replacement value for the feedback resistor Rf is calculated as shown in figure 10.
Consequently, replacing the original 10kΩ feedback resistor on U2 with a precise resistance of 4848.48 will calibrate the DAC to output exactly 3.0 V at 1,1,1,1.

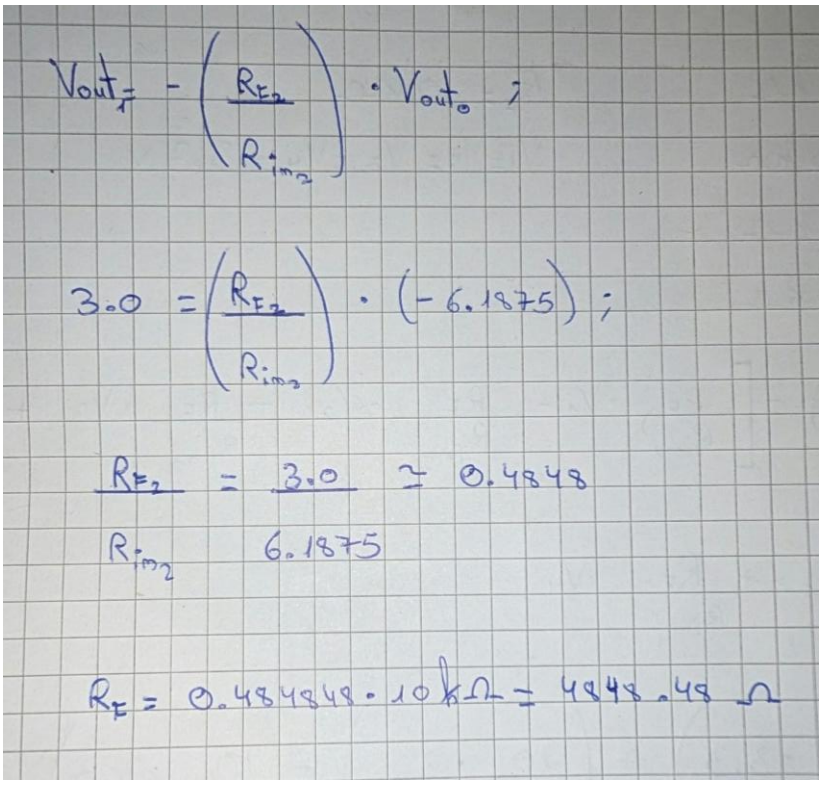


Figure 10: Gain Ratio Calculation

Part 6: Verification and Testing

To check how well the reconstruction stage worked as a whole, the final DAC output was observed on the oscilloscope and compared with the original 2 Hz sine wave from Pin P1.4, shown earlier in Figure 2 and Figure 7. At this initial 2 Hz setting, the 20 Hz timer interrupt took 10 samples per cycle. This gave the summing amplifier enough data points to follow the general shape of the input wave, even though several voltage steps were missing due to the loose breadboard connections mentioned in Part 4.

The next test checked the Nyquist limit by slowly turning up the frequency on the signal generator past 2 Hz while leaving the MSP430 sampling clock locked at 20 Hz. Once the input reached 4 Hz and 5 Hz, the microcontroller was only taking 5 and 4 samples per wave cycle. Because of this, the steps on the oscilloscope screen became much wider, and the output started looking more like a rough triangle wave than a smooth sine wave.

Following the Nyquist sampling theorem ($f_s \geq 2f_{max}$), a 20 Hz sampling rate can theoretically handle a maximum input frequency of 10 Hz, which corresponds to just two samples per cycle. During the lab test, however, the signal shape broke down before even hitting that limit, and pushing the generator past 10 Hz caused clear aliasing. Above 10 Hz, the ADC simply could not keep up with how fast the sine wave was changing. As a result, the reconstructed trace on the scope folded back on itself, showing a false, slower wave that no longer matched the actual input signal.

Part 7: Conclusion

Overall, this lab work showed how to build and test a complete 4-bit Pulse Code Modulation (PCM) setup. An incoming 0V to 3V sine wave was sampled at 20 Hz using the hardware timers on the MSP430, scaled down from the 10-bit ADC readings into a 4-bit binary format, and then rebuilt into an analogue wave through a two-stage UA741 op-amp circuit. Testing the circuit first with fixed DC voltages confirmed that the binary outputs on pins P1.0 to P1.3 matched the hand-calculated values.

Dropping the resolution from 10 bits down to 4 bits left clear staircase steps on the output wave, which is a direct example of quantisation error and noise. Looking at the op-amp maths also proved why the second stage was necessary: the first summing op-amp gave -6.1875 V at 1111, meaning the second inverting op-amp needed a gain ratio of 0.4848 (replacing the 10kΩ feedback resistor with 4848.48 Ω) to flip the voltage back to positive and keep the peak at 3.0 V. In addition, sweeping the input frequency above 2 Hz backed up the Nyquist theorem in practice, showing that any signal approaching or passing half the sampling rate (10 Hz) gets ruined by aliasing.

Putting the binary-weighted DAC together on a breadboard also pointed out some clear hardware drawbacks. Because this circuit depends entirely on doubling the resistor values for each bit (10 kΩ, 20 kΩ, 40 kΩ, and 80 kΩ), small resistor tolerances or a bad contact in the breadboard tracks can easily throw off the output. That was seen firsthand when the reconstructed wave dropped from the expected 16 steps down to around 8 steps. Also, if a system needed 8 or 10 bits of resolution, the resistor values in this type of DAC would become way too large to be practical.

Finally, it is worth noting that this lab used a 4-bit parallel output rather than the serial connections used in real-world telecom systems. Sending all four bits at once over four separate wires made the breadboard circuit much simpler, since the op-amp could add the voltages right away without needing shift registers or clock recovery. However, parallel wiring is only practical over short distances on a bench, for long-distance lines, serial PCM is always preferred to save wire costs, allow Time Division Multiplexing (TDM), and avoid timing skew between separate cables.